# D4.3 Climate change (HL)

| NAME | DATE | TOTAL |
|---|---|---|
| | | _____ / 30 |

**About 30 minutes,** or one section at a time. The number of marks shows how many separate points to make.

## Section A: Recall

6 MARKS

1 **DEFINE** phenology. [D4.3.9] [1]

2 **STATE** two variables that influence the timing of biological events. [D4.3.9] [2]

3 **STATE** one phenological event in a deciduous tree. [D4.3.9] [1]

4 **NAME** an insect that has more generations per year because of warming. [D4.3.11] [1]

5 **STATE** which colour of tawny owl has become more common in Finland. [D4.3.12] [1]

## Section B: Understanding

11 MARKS

6 **EXPLAIN** how climate change can cause great tit chicks to hatch at the wrong time. [D4.3.10] [3]

7 **EXPLAIN** how an increase in bark beetle generations threatens spruce forests. [D4.3.11] [3]

8 **OUTLINE** how warming affects reindeer through their food plants. [D4.3.10] [2]

**9** **EXPLAIN** how reduced snow cover has changed the frequency of colour variants in tawny owls. [D4.3.12] [3]

## Section C: Working with evidence

9 MARKS

*All data in this section are invented for practice. The patterns are realistic.*

**10** [D4.3.10]

In a forest, the dates of two events were recorded. Dates are given as days of the year.

| YEAR | 1980 | 1990 | 2000 | 2010 | 2020 |
|---|---|---|---|---|---|
| PEAK NUMBER OF CATERPILLARS | 150 | 146 | 141 | 137 | 132 |
| PEAK FOOD DEMAND OF GREAT TIT CHICKS | 150 | 148 | 146 | 144 | 142 |

a **CALCULATE** the mean change per decade in the date of peak caterpillar numbers. [2]

b **CALCULATE** the mismatch between the two events in 2020. [1]

c **SUGGEST** why the dates of the two events did not change equally. [2]

d **PREDICT** the effect of this mismatch on the great tit population. [2]

e **EXPLAIN** how natural selection could reduce the mismatch. [2]

## Section D: Longer answer

4 MARKS

**11** **EXPLAIN** how a longer growing season could cause the date of bud set in Scots pine to evolve. [D4.3.12] [4]